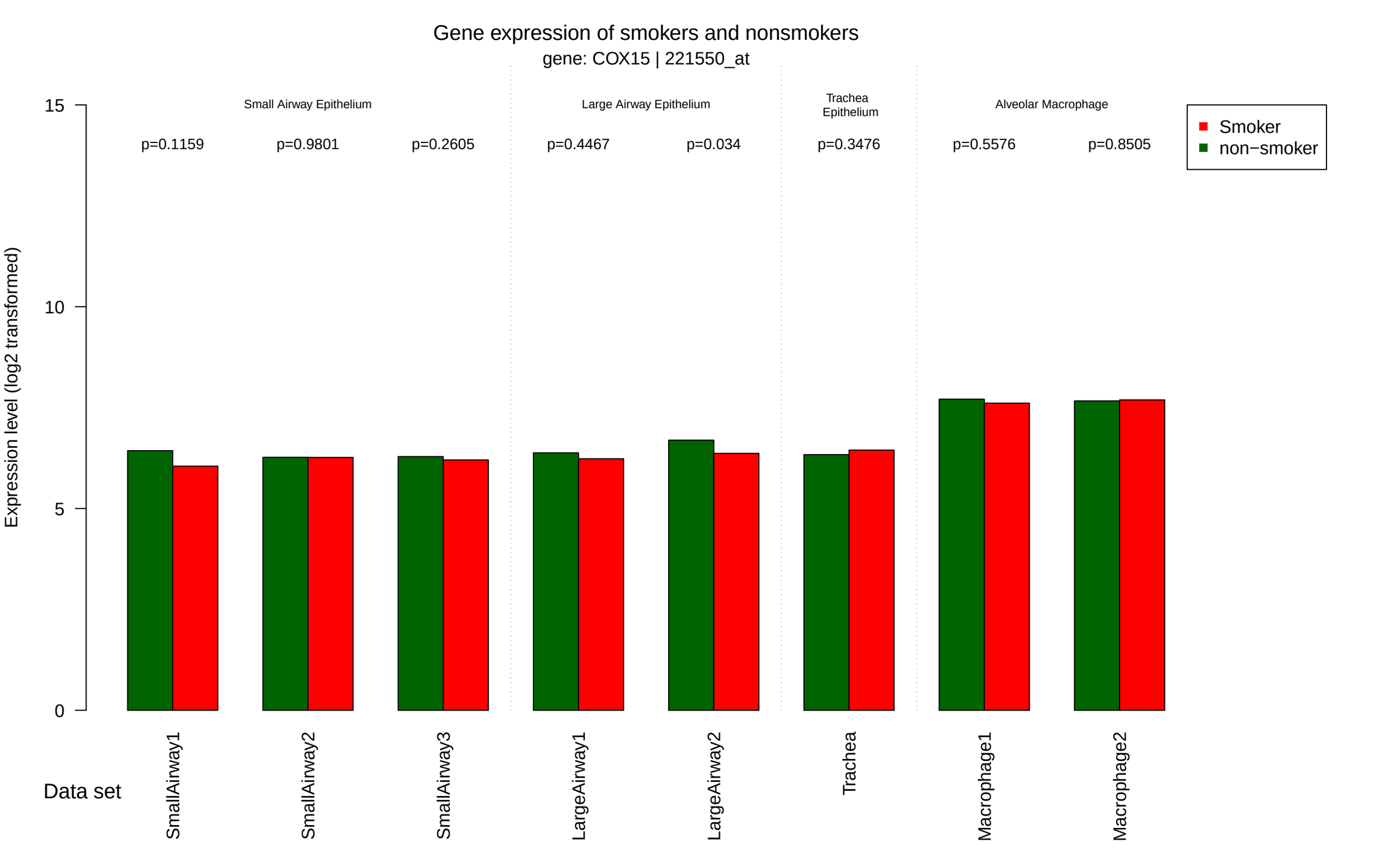
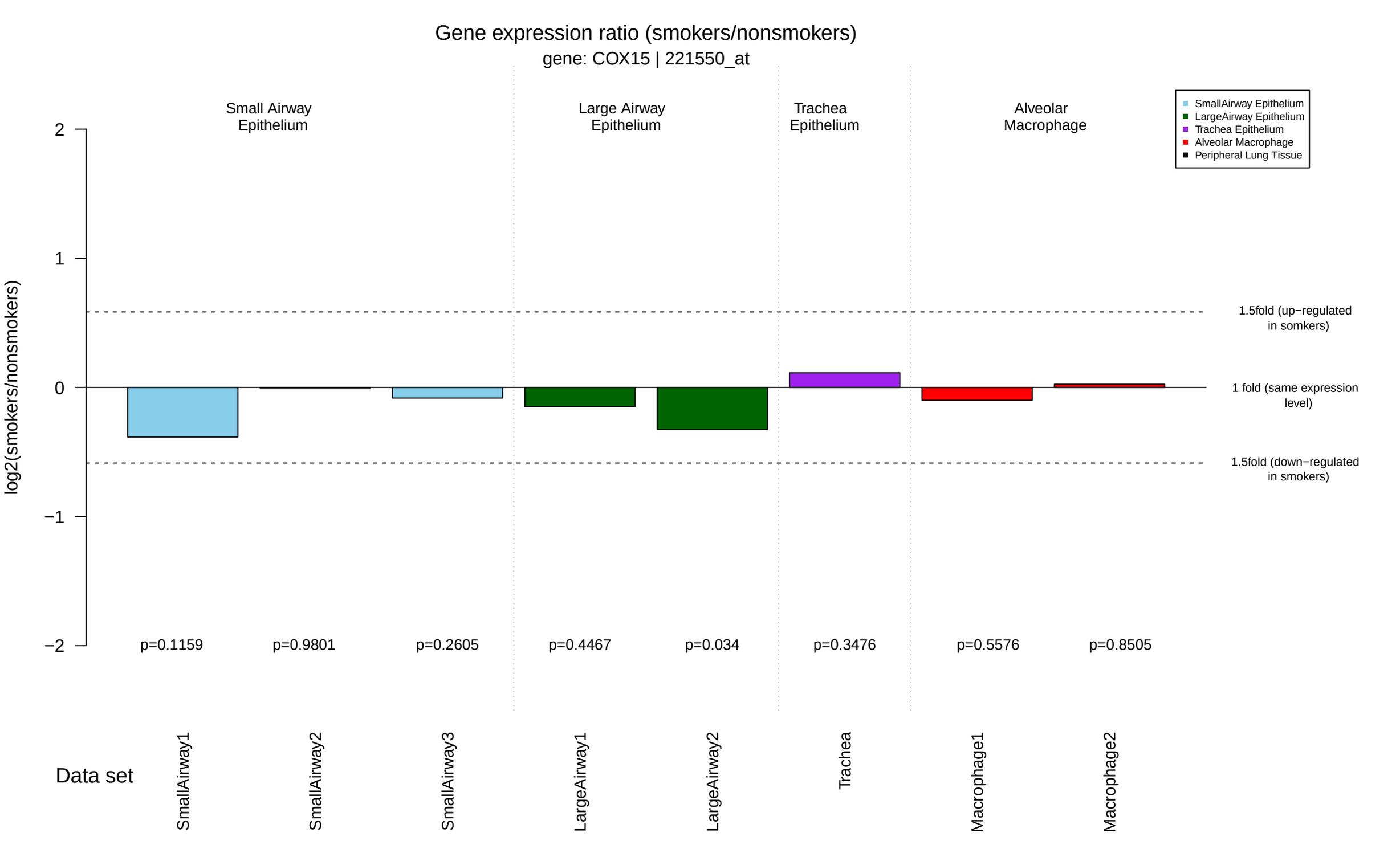
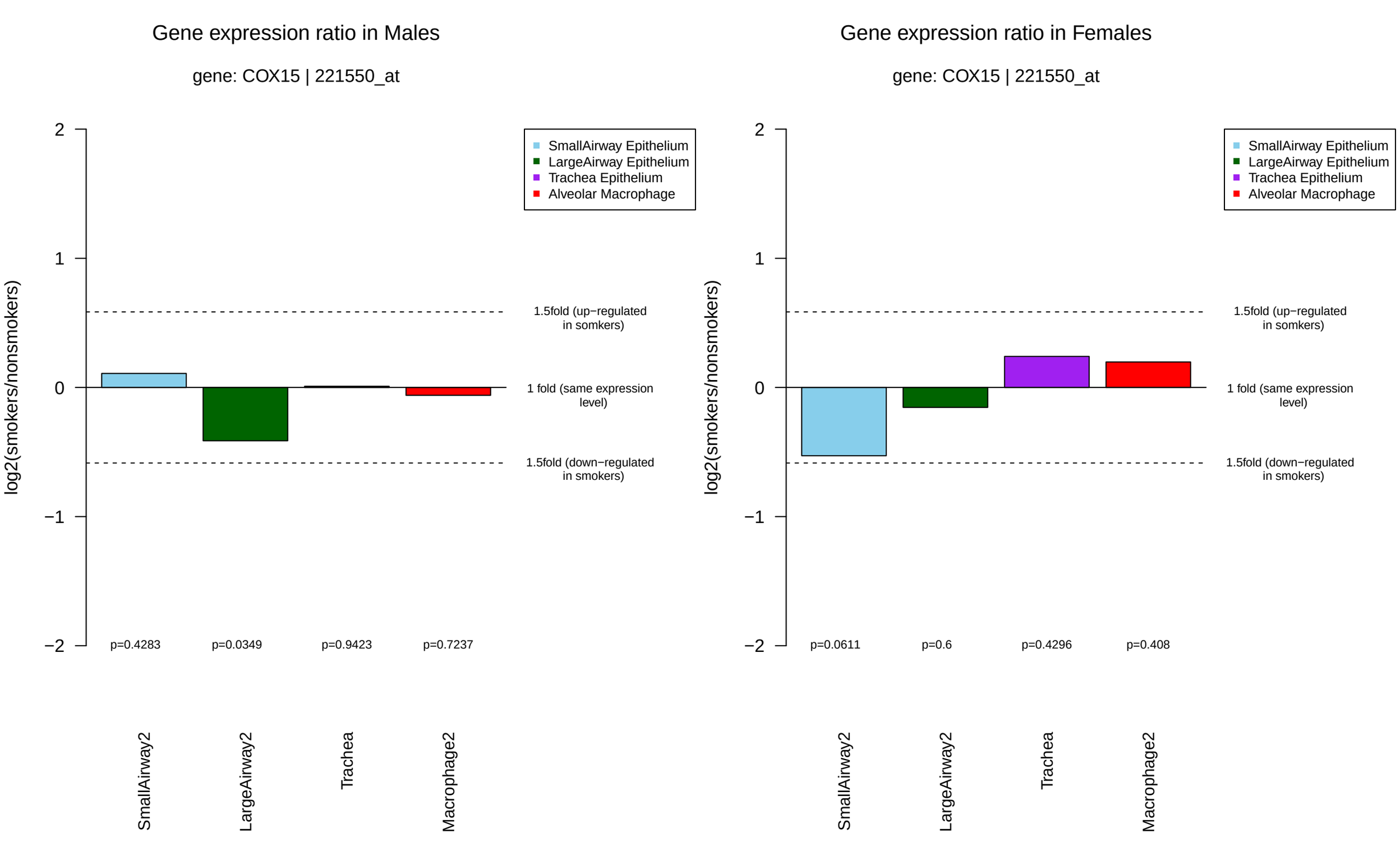
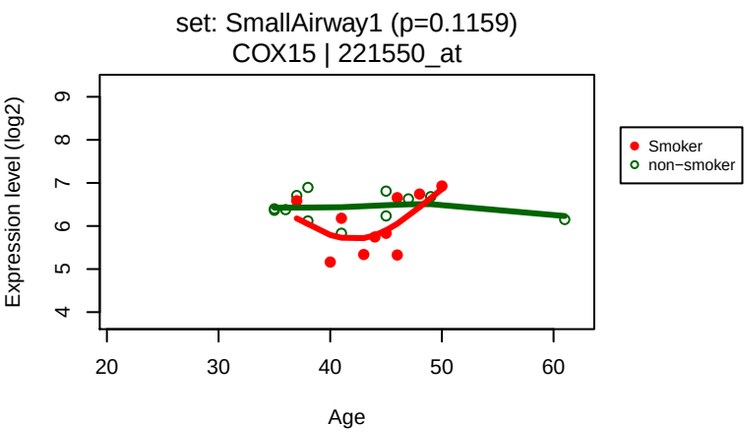
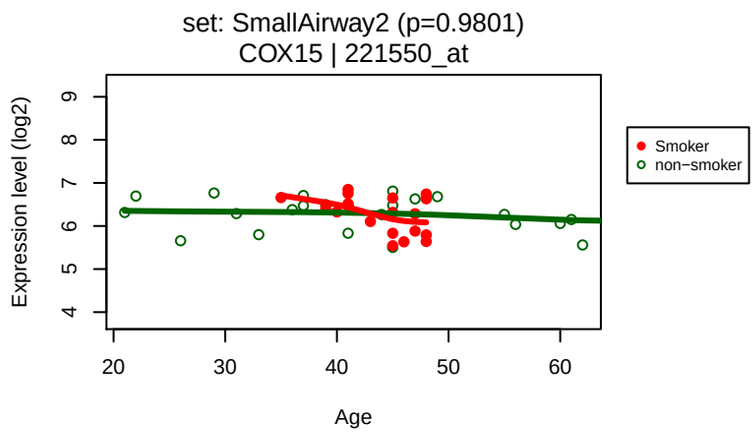
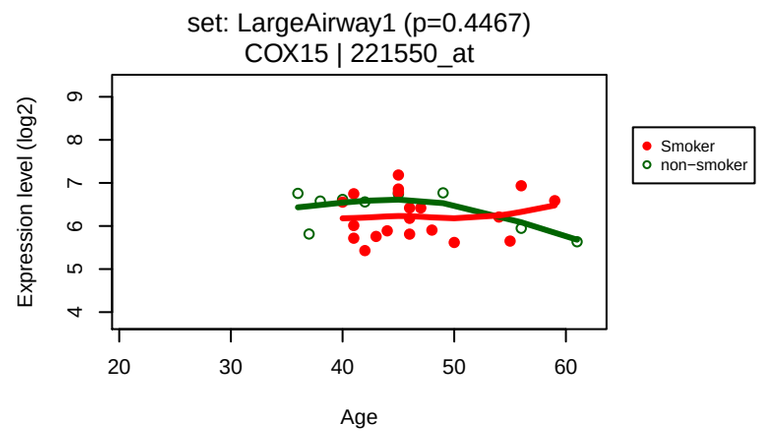

Gene expression of smokers and nonsmokers
gene: COX15 | 221550_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1159
p=0.9801
p=0.2605
p=0.4467
p=0.034
p=0.3476
p=0.5576
p=0.8505
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

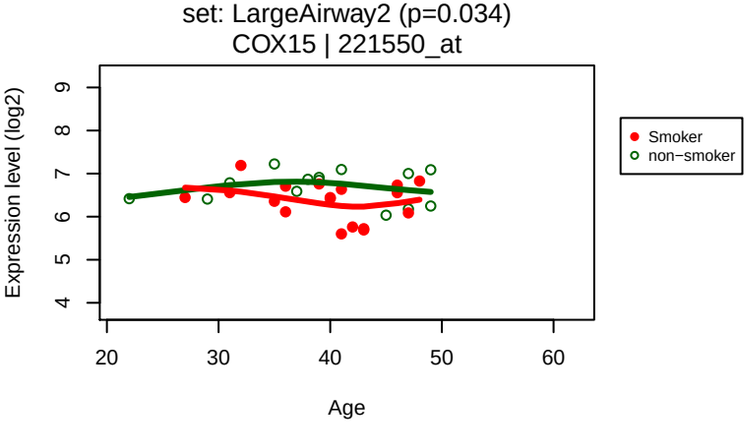

Gene expression ratio (smokers/nonsmokers)
gene: COX15 | 221550_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1159
p=0.9801
p=0.2605
p=0.4467
p=0.034
p=0.3476
p=0.5576
p=0.8505
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

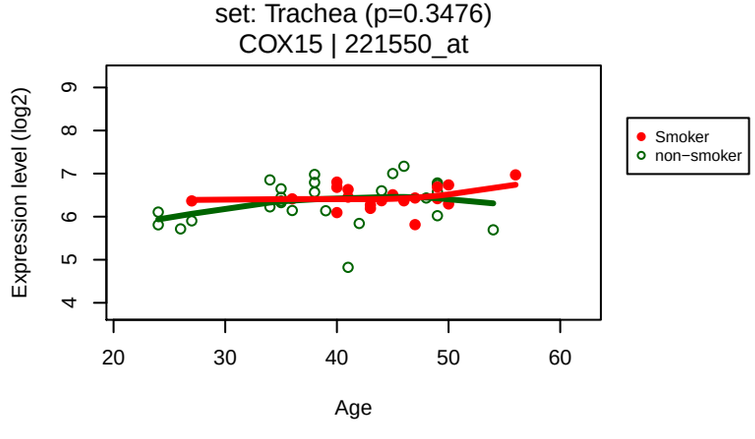

Gene expression ratio in Males
gene: COX15 | 221550_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4283
p=0.0349
p=0.9423
p=0.7237
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COX15 | 221550_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0611
p=0.6
p=0.4296
p=0.408
SmallAirway2
LargeAirway2
Trachea
Macrophage2

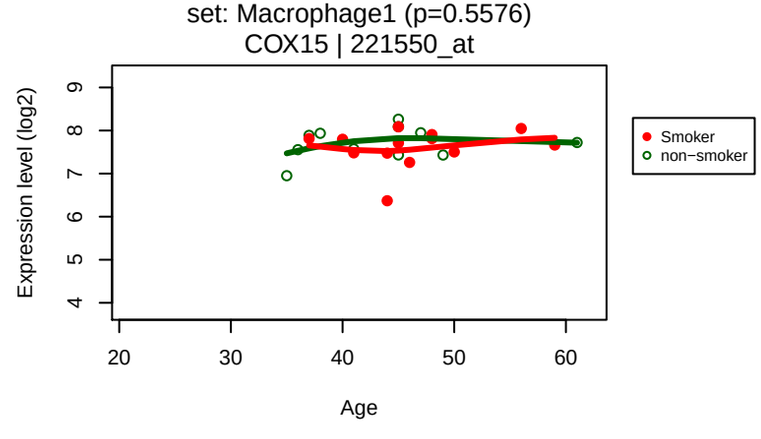

set: SmallAirway1 (p=0.1159)
COX15 | 221550_at
Expression level (log2)
Age
Smoker
non−smoker

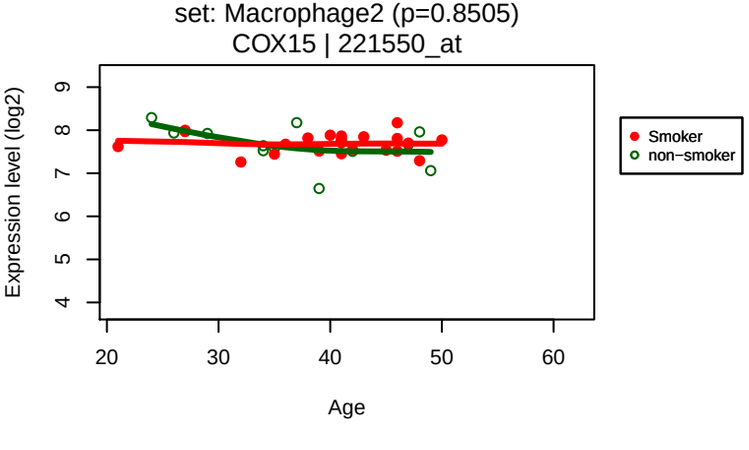

set: SmallAirway2 (p=0.9801)
COX15 | 221550_at
Expression level (log2)
Age
Smoker
non−smoker

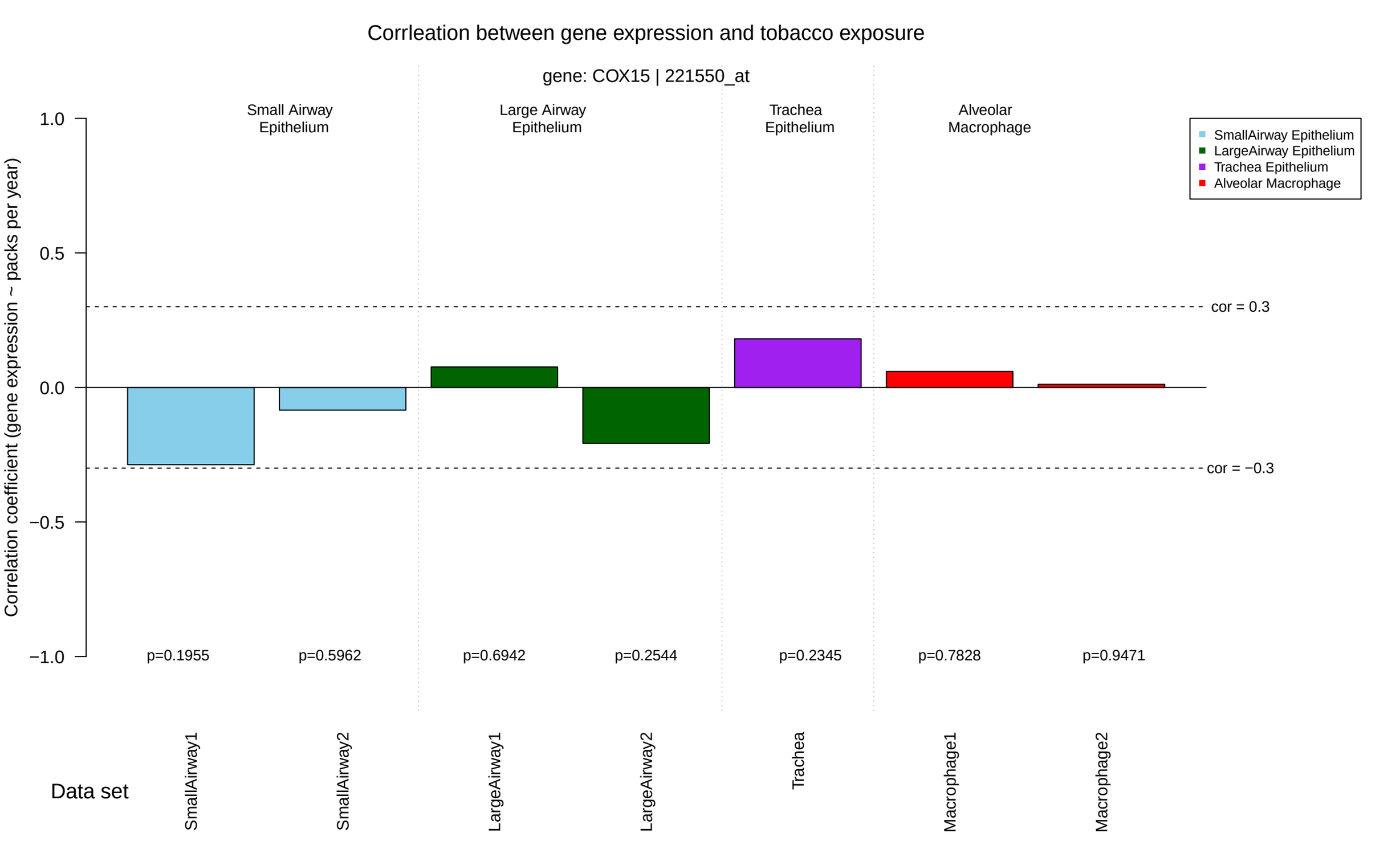

set: LargeAirway1 (p=0.4467)
COX15 | 221550_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.034)
COX15 | 221550_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3476)
COX15 | 221550_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5576)
COX15 | 221550_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8505)
COX15 | 221550_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COX15 | 221550_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1955
p=0.5962
p=0.6942
p=0.2544
p=0.2345
p=0.7828
p=0.9471
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2